

Green Hydrogen Pilot – Feasibility Study

5 MW PEM electrolyser at Port of Eastholm with Arcturus Steel AG · Final report

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Executive summary

DRAFT – LCOH figures preliminary, vendor data from two suppliers outstanding. Base case LCOH currently EUR 10.40/kg without grant (capex EUR 22.8 m incl. tube trailer logistics instead of pipeline).

The pilot is technically feasible and has no safety showstoppers. In the base case it produces **481 t of hydrogen per year** at a levelised cost of hydrogen (LCOH) of **EUR 9.72/kg**. With an EU Innovation Fund grant of EUR 10.5 m the LCOH falls to **EUR 7.66/kg**, below the EUR 8/kg threshold set at the go/no-go gate in January 2026. The study recommends proceeding to the funding application and to FEED.

Key figure	Value
Electrolyser	5 MW PEM (two 2.5 MW stacks)
Hydrogen output	481 t/a (≈ 1,319 kg/day average)
Full-load hours	5,200 h/a
Specific consumption (system)	54.0 kWh/kg
Capex	EUR 21.5 m
LCOH without grant	EUR 9.72/kg
LCOH with Innovation Fund grant	EUR 7.66/kg
CO2 avoided at Arcturus	≈ 3.2 kt/a (replacing natural gas in the furnace)
Target commercial operation	Q4 2028

Table 1 – Pilot at a glance



Concept rendering of the electrolyser plant on the Eastholm quayside

1. Background and objectives

Arcturus Steel AG is the offtaker of Weyland's 15-year PPA WE-PPA-2026-004 (150 MW from Harrow Bay, signed 12 May 2026). Arcturus wants to decarbonise its rolling mill at Eastholm, where natural gas fires the reheating furnace. Hydrogen co-firing at up to 30 % (by energy) is possible with burner modifications; the pilot is sized to supply this trial phase and later a full furnace zone.

Objectives of the study: (1) confirm technical feasibility at the port site, (2) estimate LCOH and funding need, (3) select electrolyser technology, (4) identify safety and permitting requirements, and (5) define the commercial structure with Arcturus. The study team was led by Isabel Ferreira (Hydrogen Project Engineer) under the steering of Sofie Lindqvist and an Arcturus energy manager.

2. Site

The site is a 0.9 ha gravel plot on the eastern quay of the Port of Eastholm, next to the planned Harrow Bay O&M base. It has a 33 kV supply nearby, process water from the municipal network and space for a 1.4 km pipeline along the existing port service corridor to the Arcturus rolling mill.

IMAGE PLACEHOLDER

hydrogen-port-aerial

Aerial drone photograph of a mid-sized North Sea port with quays, warehouses, a few cargo ships and an empty gravel plot next to the water, a large steelworks rolling mill visible in the background, overcast daylight.

Port of Eastholm – eastern quay plot with the Arcturus rolling mill in the background

3. Hydrogen demand

Arcturus plans to use about 330 t in the trial year and around 490 t/a from year three, with lower demand in July (mill maintenance shutdown). The base case production of 481 t/a matches full operation; during the trial year surplus hydrogen will be sold to a local bus operator via tube trailer (letter of intent expected Q4 2026).

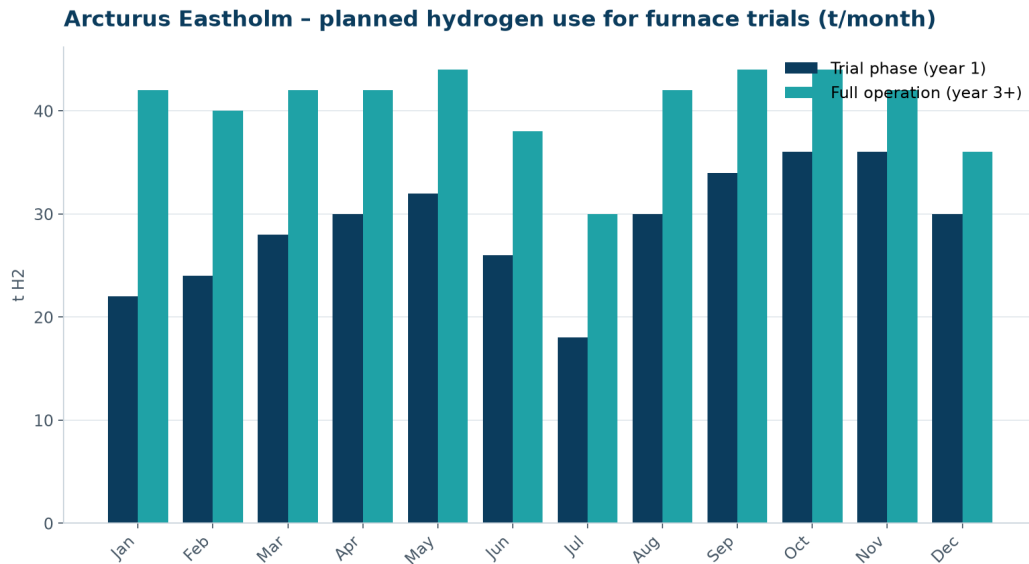


Figure 1 – Arcturus monthly hydrogen demand

4. Technology selection

Four suppliers responded to the request for information (three PEM, one alkaline). PEM was selected for its dynamic response to variable wind power, compact footprint and higher outlet pressure (30 bar). The vendor comparison is documented separately; two PEM vendors are shortlisted for FEED.

Criterion	PEM	Alkaline (pressurised)
Load range and ramp rate	5–100 %, seconds	15–100 %, minutes
Specific consumption (system, BoL)	53–56 kWh/kg	51–54 kWh/kg
Footprint for 5 MW	≈ 450 m ²	≈ 750 m ²
Stack lifetime	60,000–80,000 h	80,000–95,000 h
Critical materials	Iridium, platinum	Nickel
Capex (stack + power electronics)	Higher	Lower

5. Costs

Capex item	EUR m
Electrolyser stacks and power electronics	7.4
Balance of plant (water treatment, cooling, purification)	3.1
Compression (30 → 200 bar) and buffer storage 2.5 t	2.3
Hydrogen pipeline to Arcturus rolling mill (1.4 km)	1.3
Grid connection 33 kV and transformer	1.6
Civil works and electrolyser hall	2.2
Engineering, project management, owner's costs	1.9
Contingency (≈ 9 %)	1.7
Total capex	21.5

Table 2 – Capex estimate (AACE class 4, ±30 %)

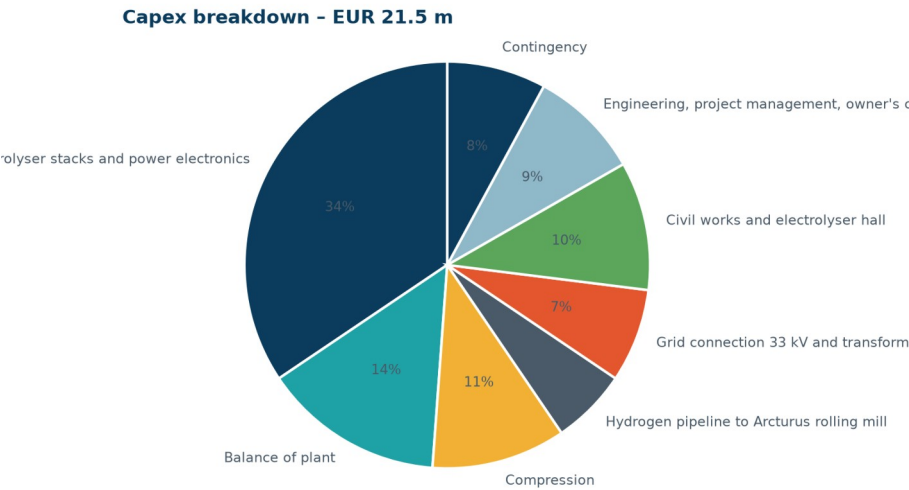


Figure 2 – Capex breakdown